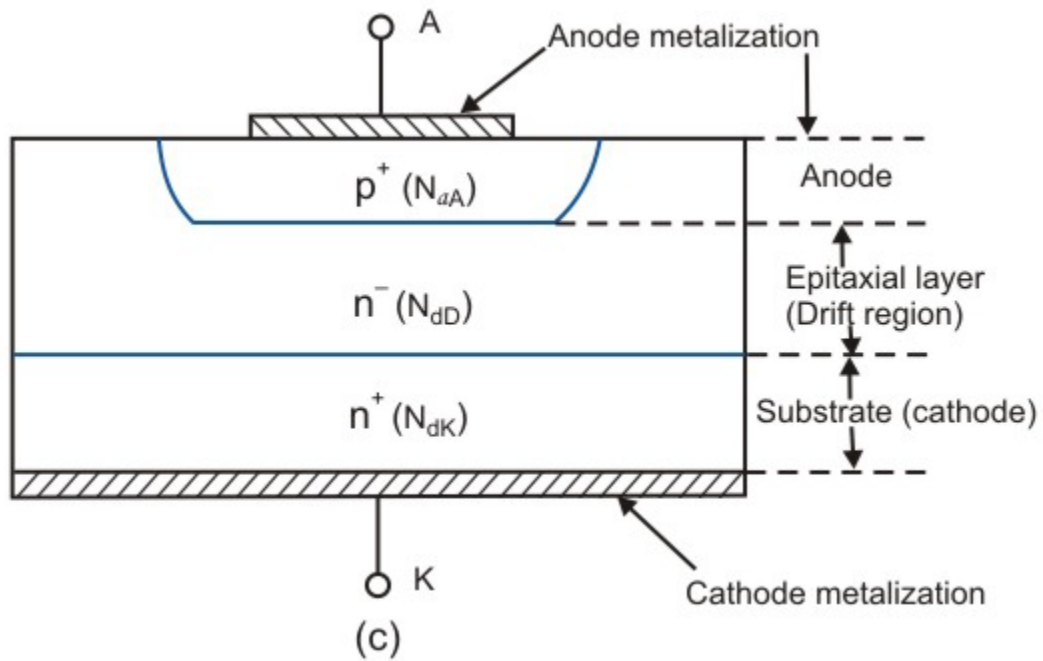
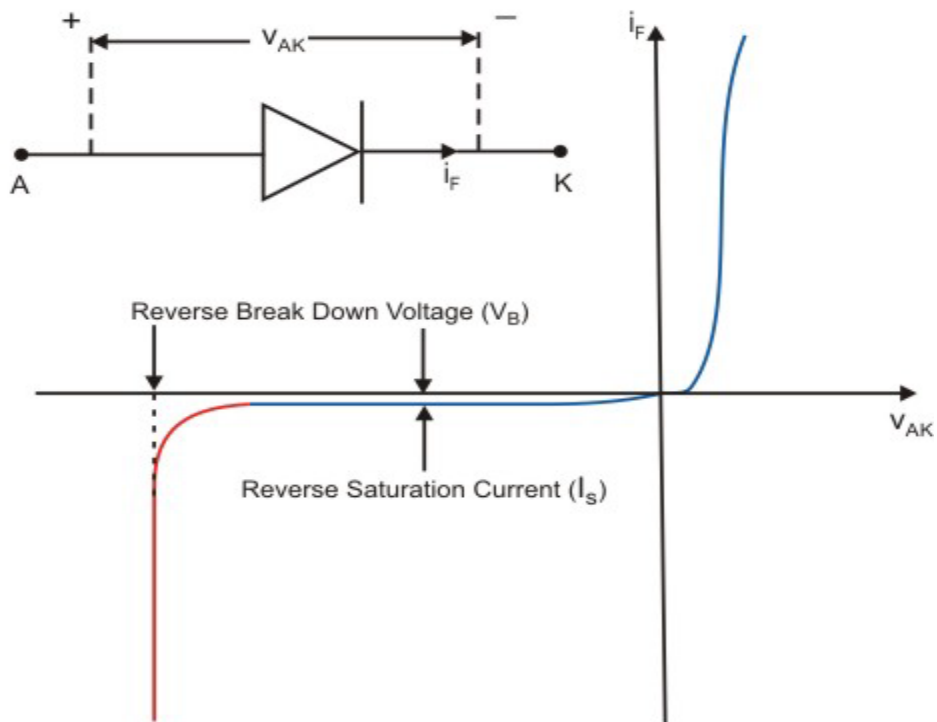
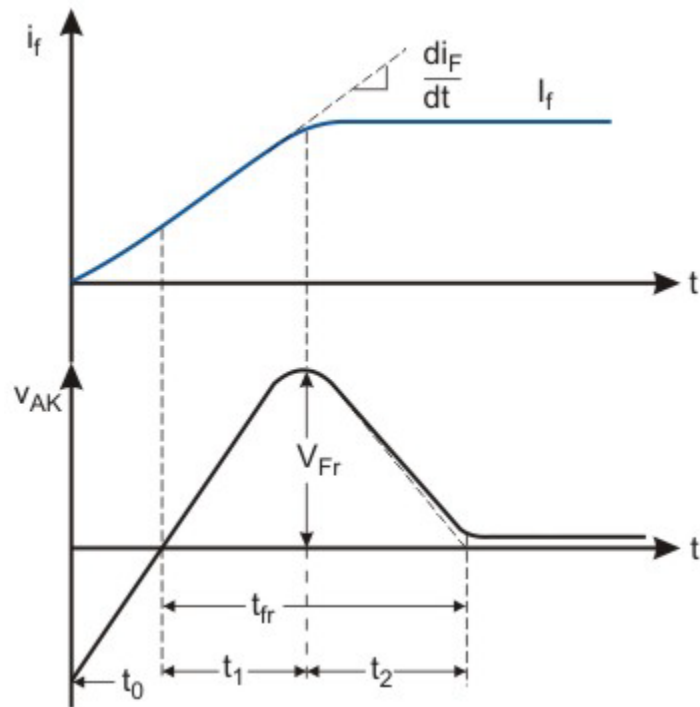


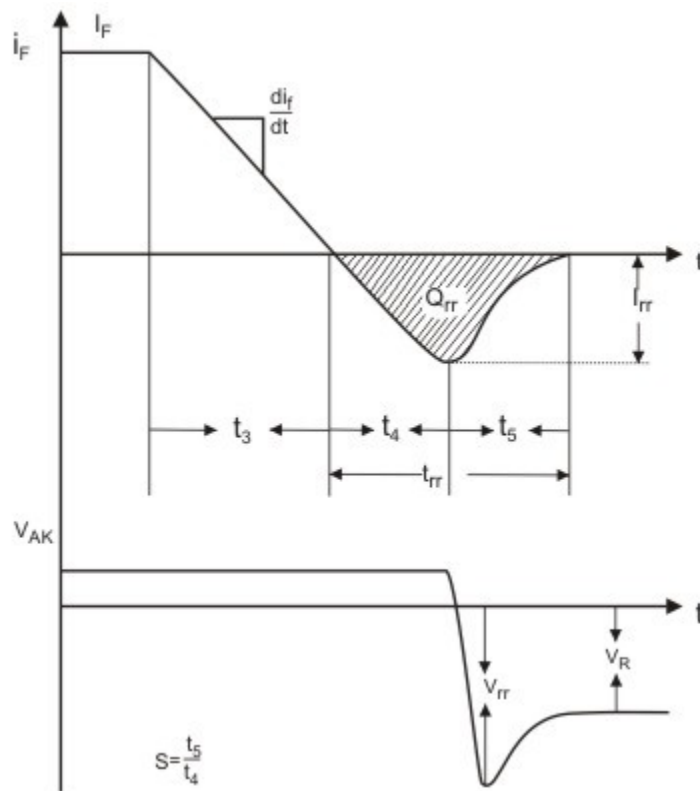
Diagram of a power diode (c) schematic cross section (a) circuit symbol.



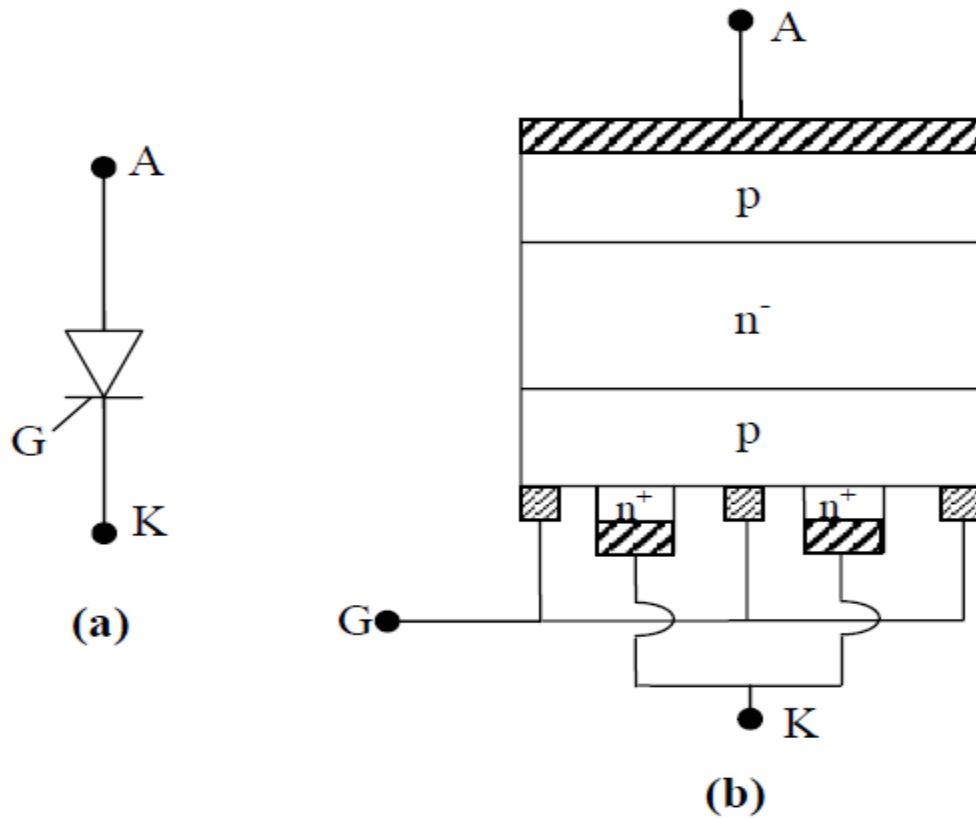
Volt-Ampere (i-v) characteristics of a p-n junction diode



Forward current and voltage waveforms of a power diode during Turn On operation.



Reverse Recovery characteristics of a power diode



Constructional features of a thyristor (a) Circuit Symbol, (b) Schematic Construction

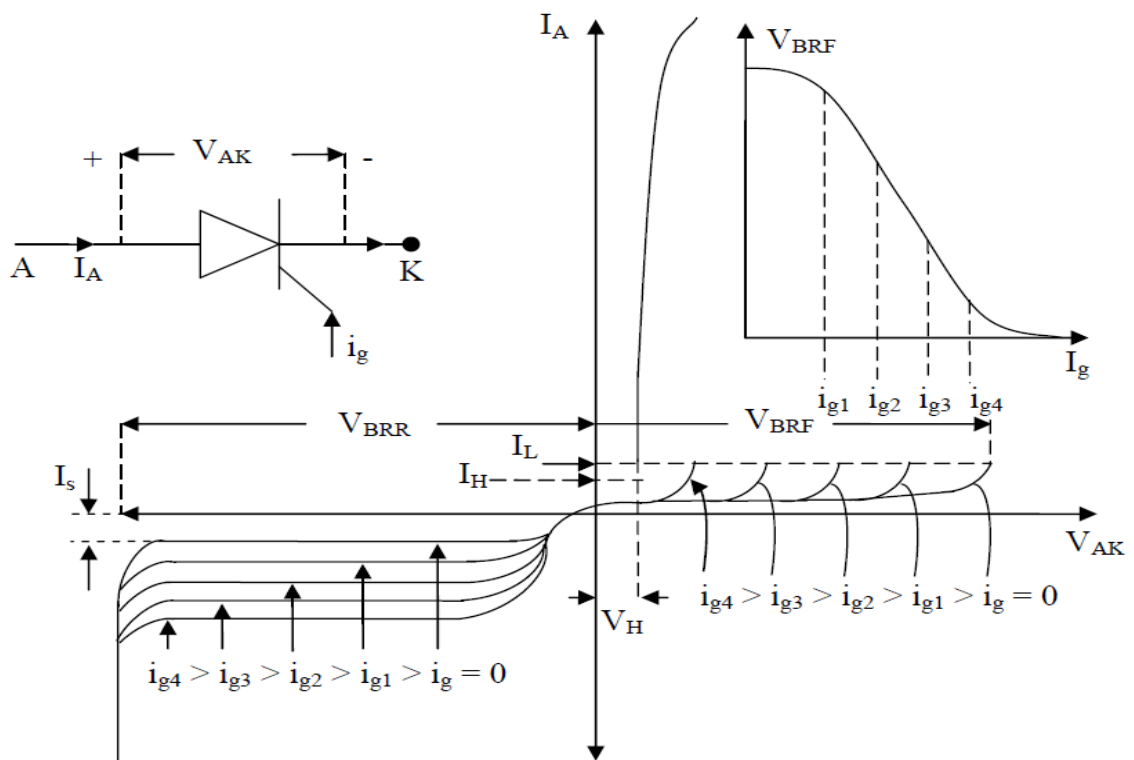
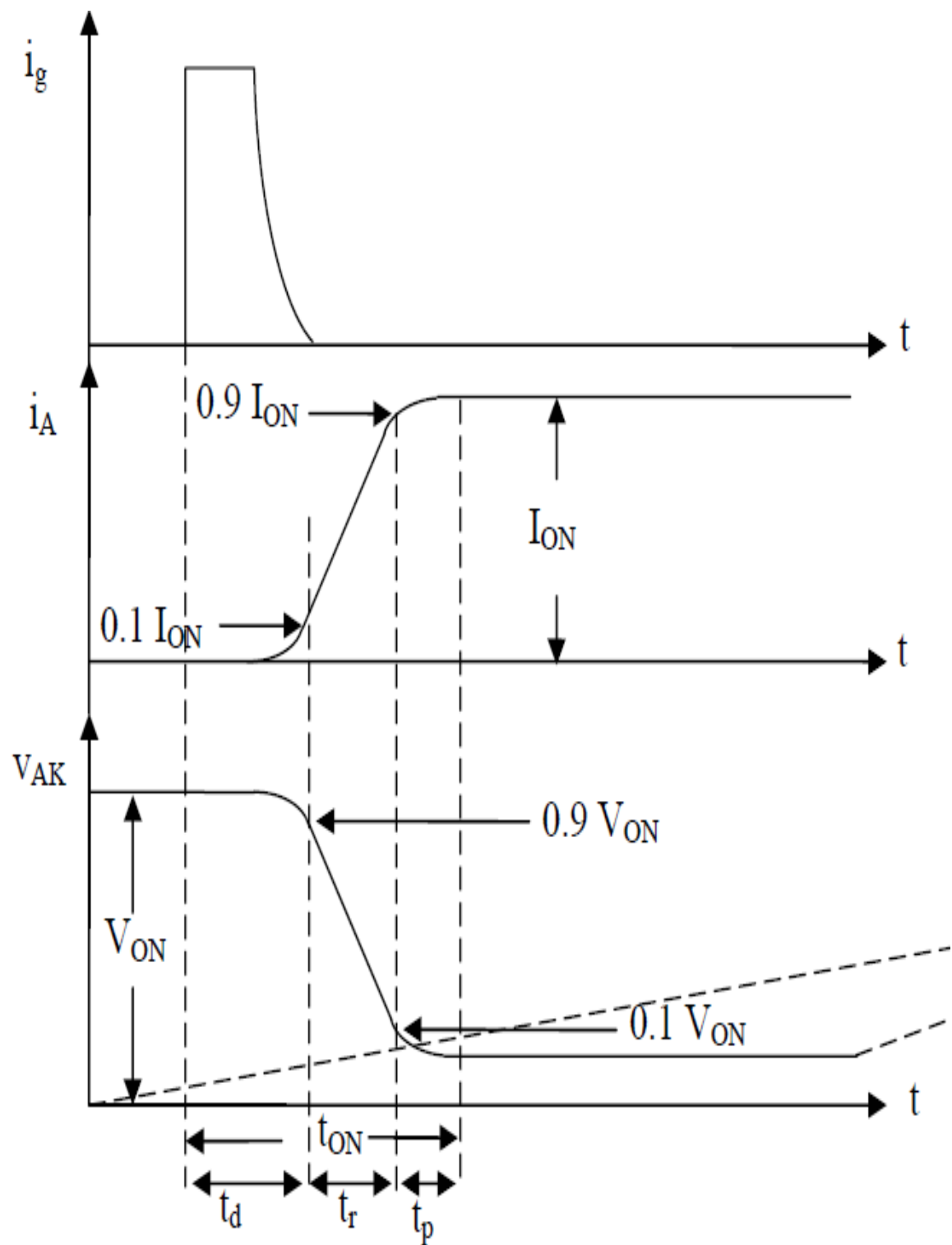
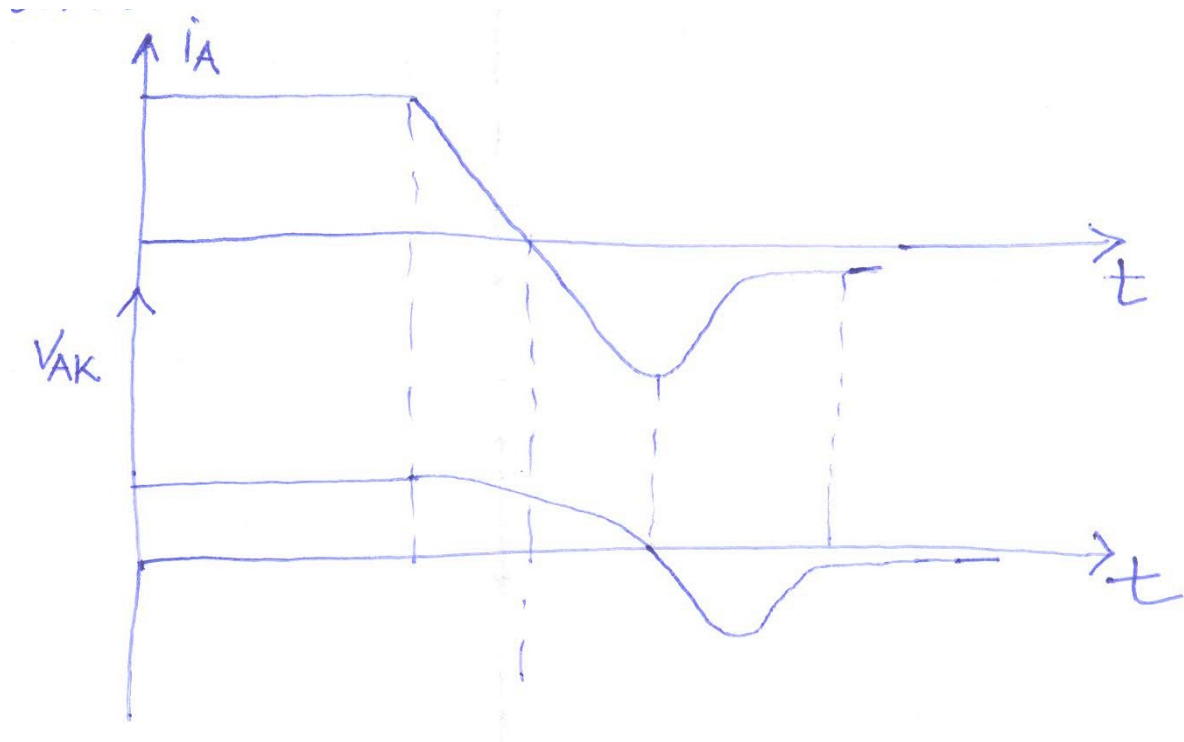


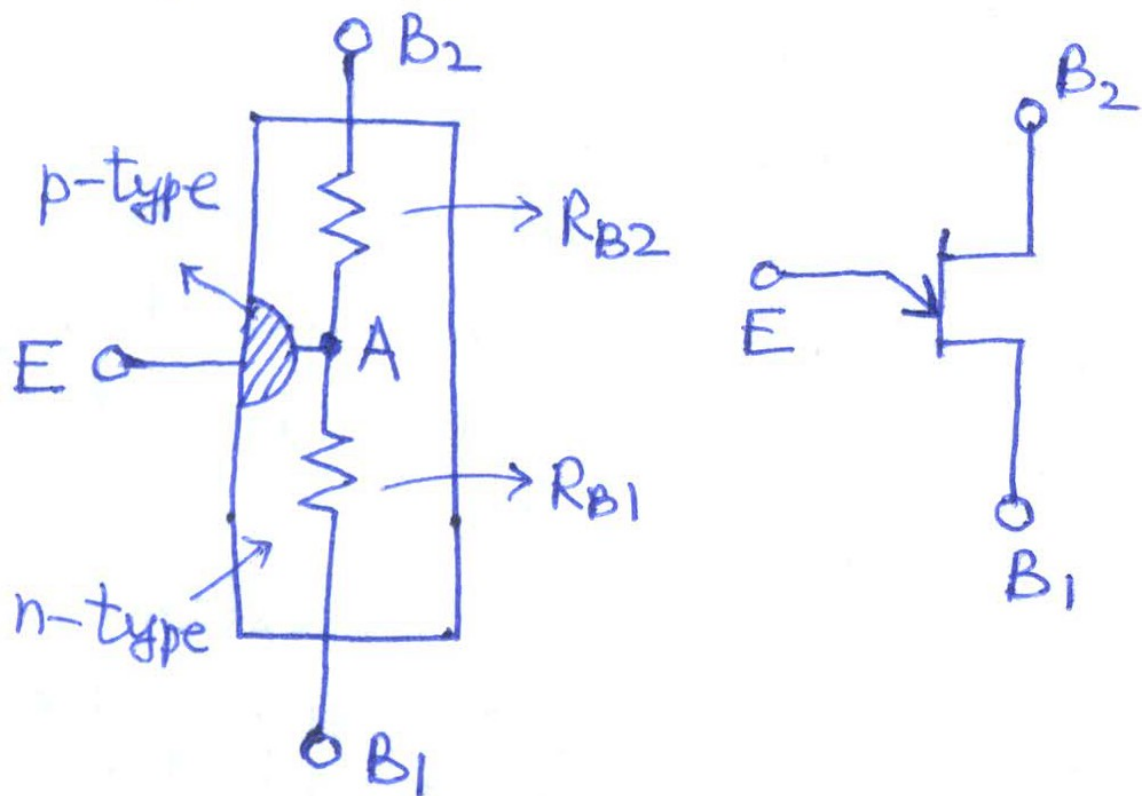
Fig. 4.3: Static output characteristics of a Thyristor



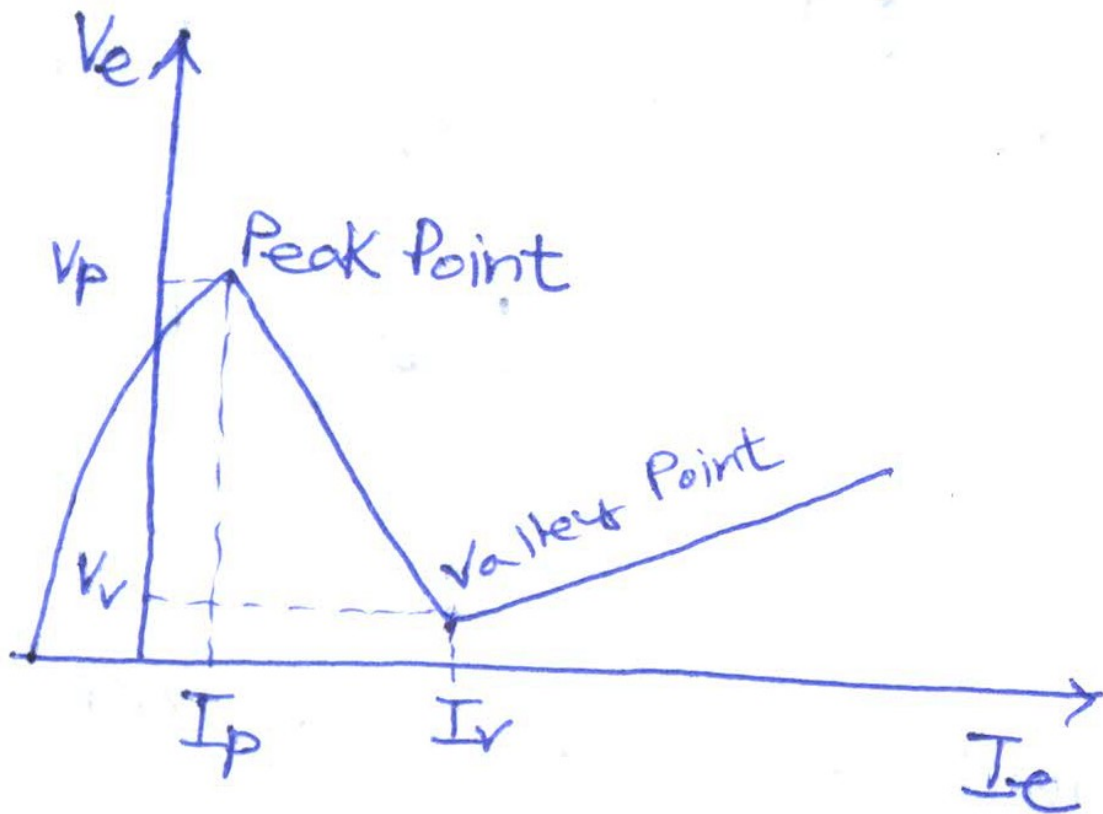
Turn on Switching Characteristics of a Thyristor



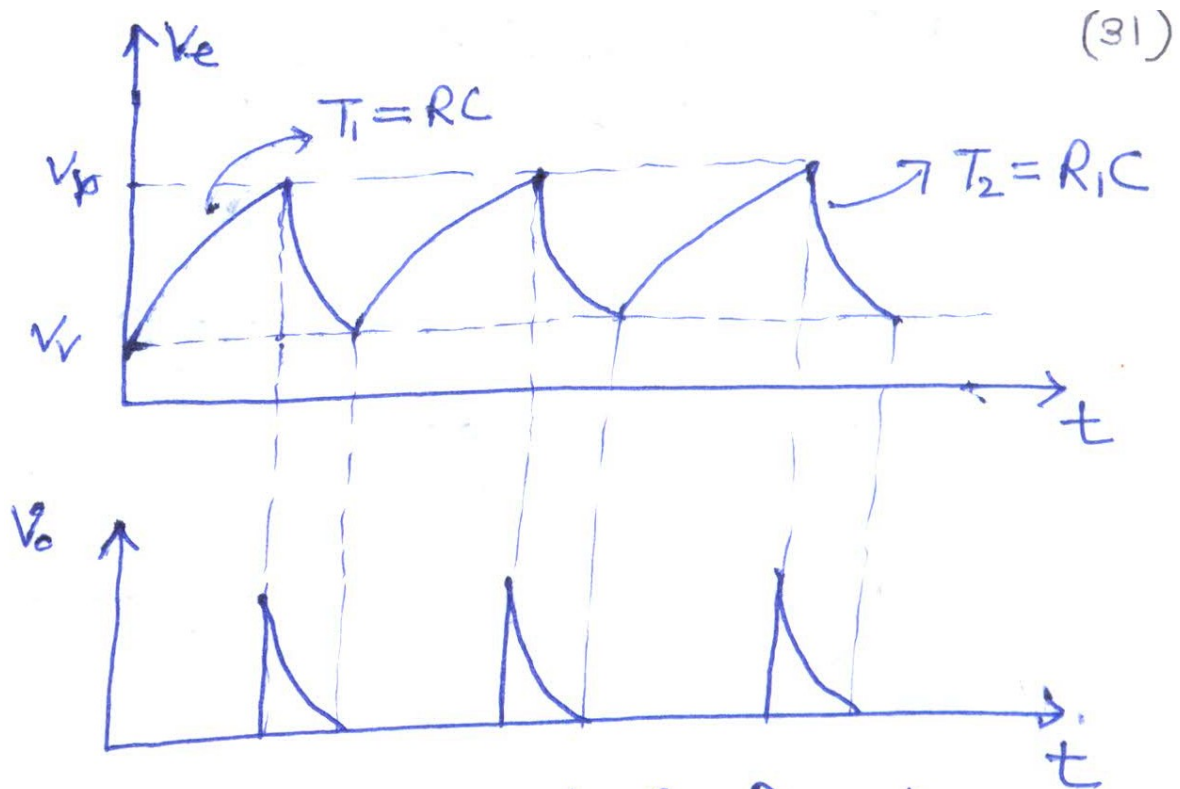
Turn off Switching Characteristics of a Thyristor



UJT Construction and Symbol



V-I characteristics of UJT



UJT Oscillator Voltage Waveforms

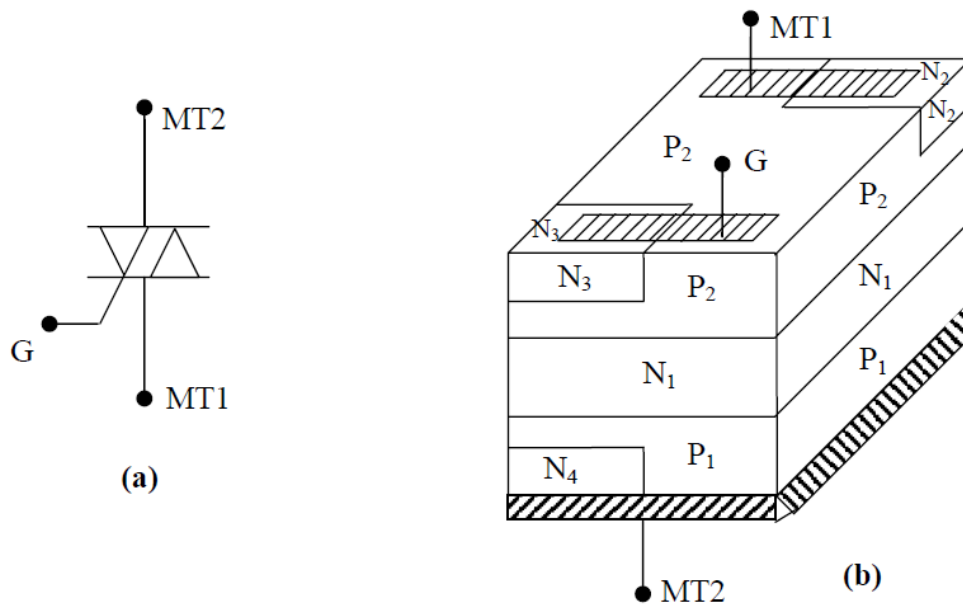


Fig. 4.12: Circuit symbol and schematic construction of a Triac
 (a) Circuit symbol (b) Schematic construction.

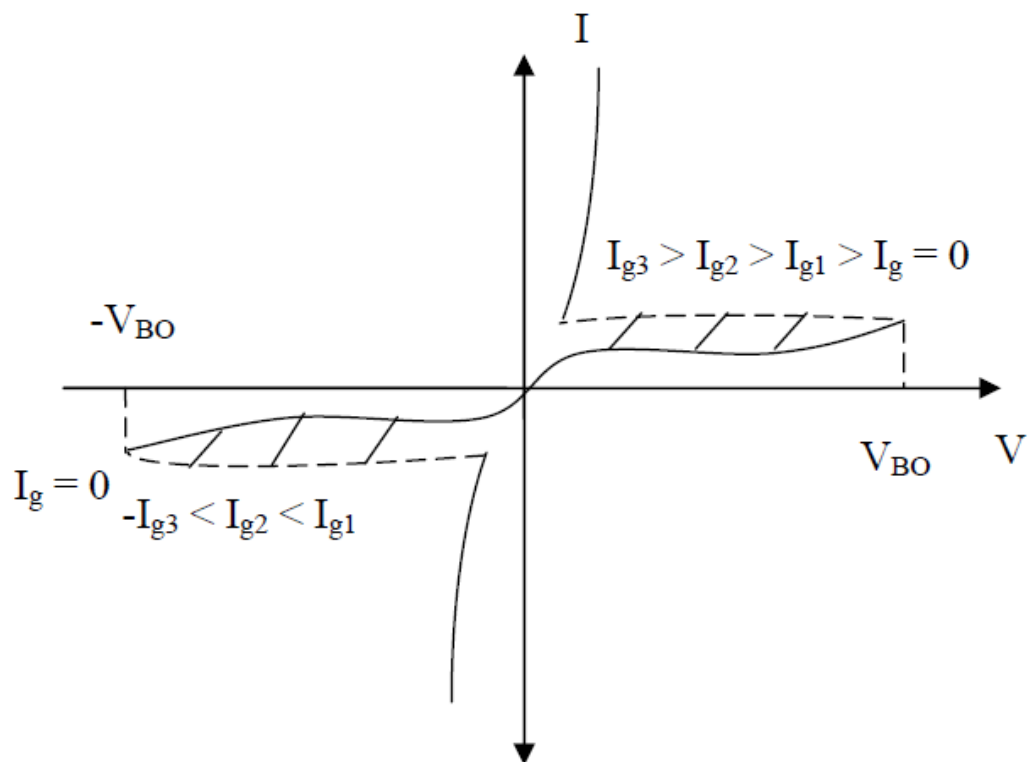
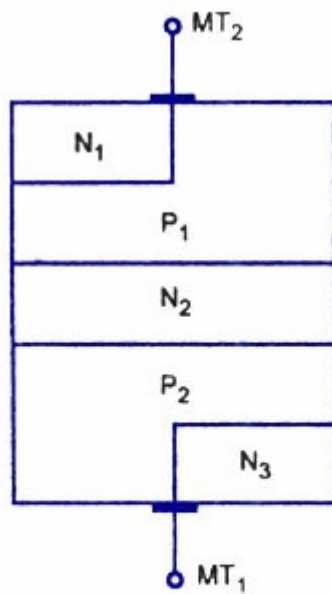
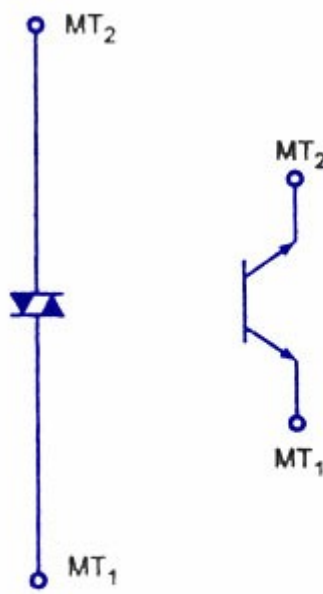


Fig. 4.14: Steady state V – I characteristics of a Triac

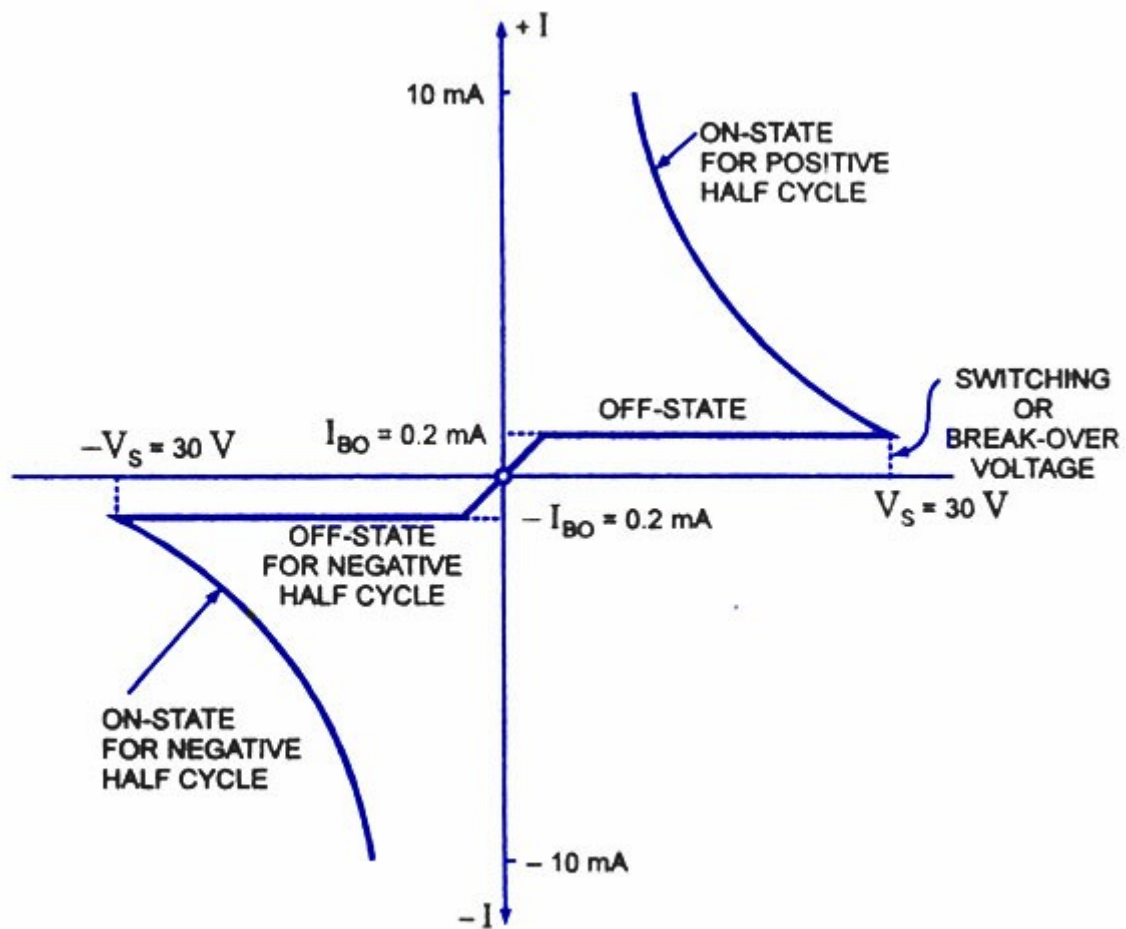


Basic Structure



Schematic Symbols

Diac



V-I Characteristic of a Diac

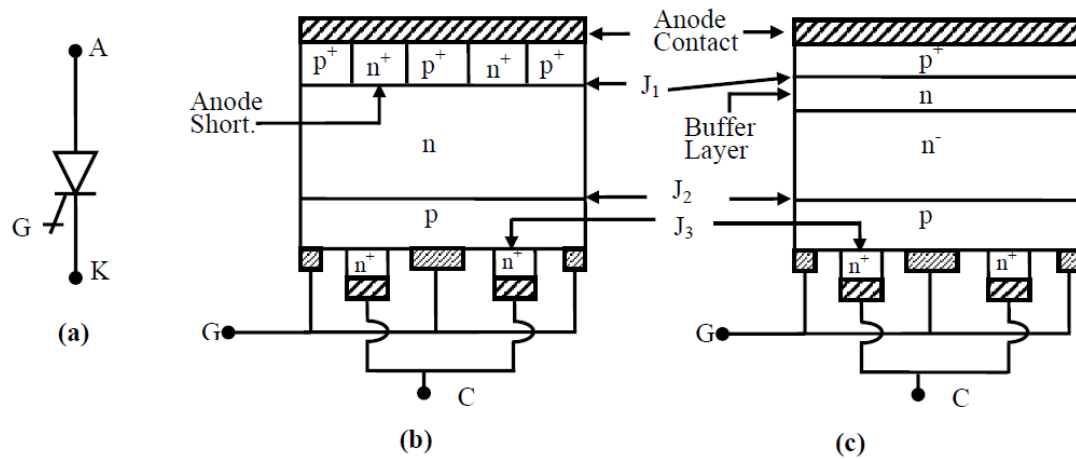


Fig. 5.1: Circuit symbol and schematic cross section of a GTO
 (a) Circuit Symbol, (b) Anode shorted GTO structure,
 (c) Buffer layer GTO structure.

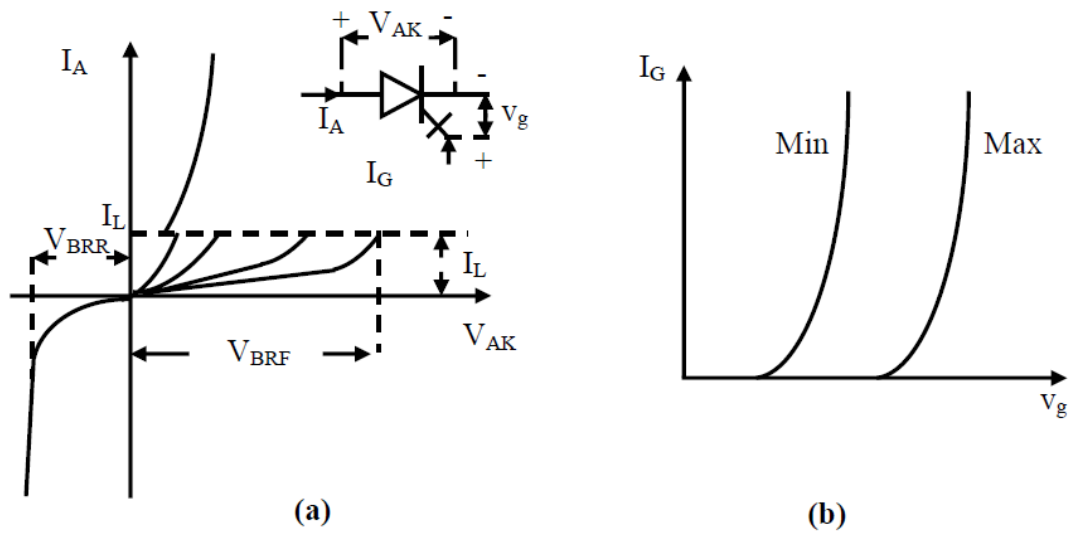
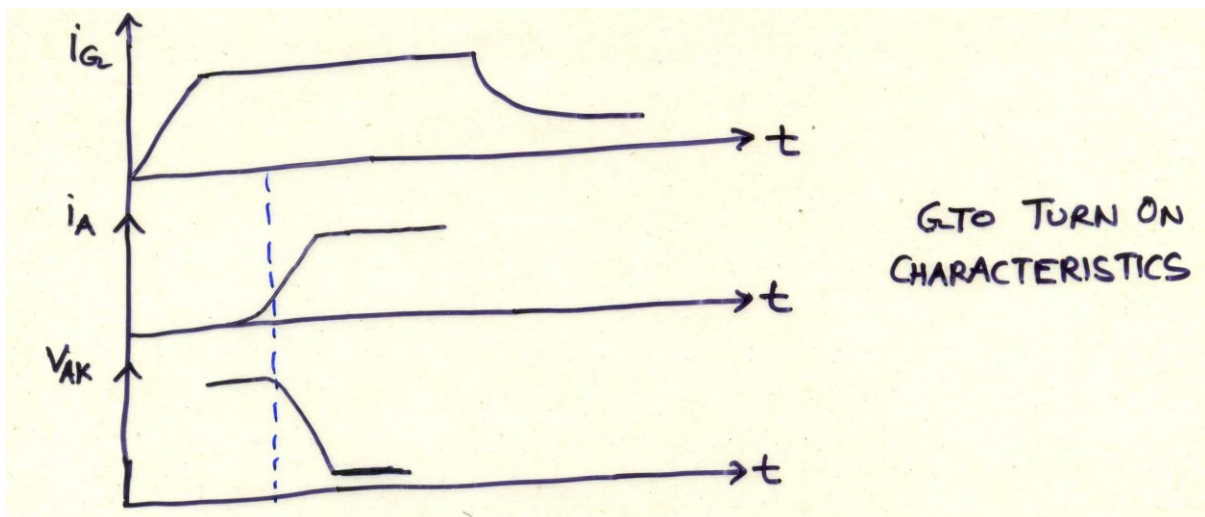
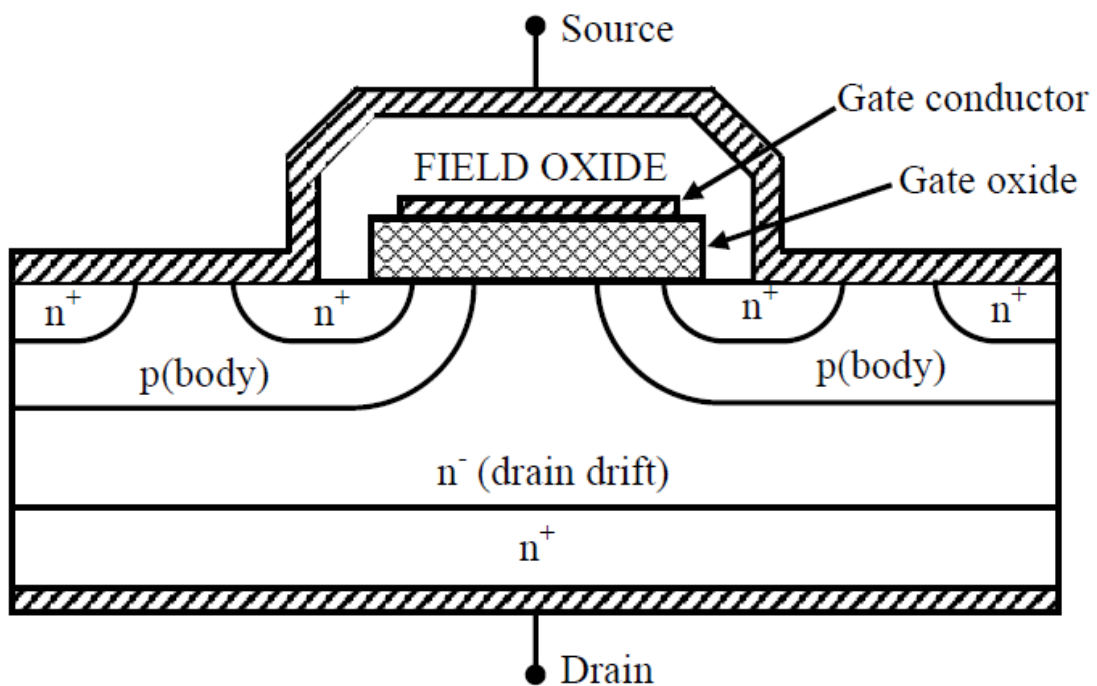
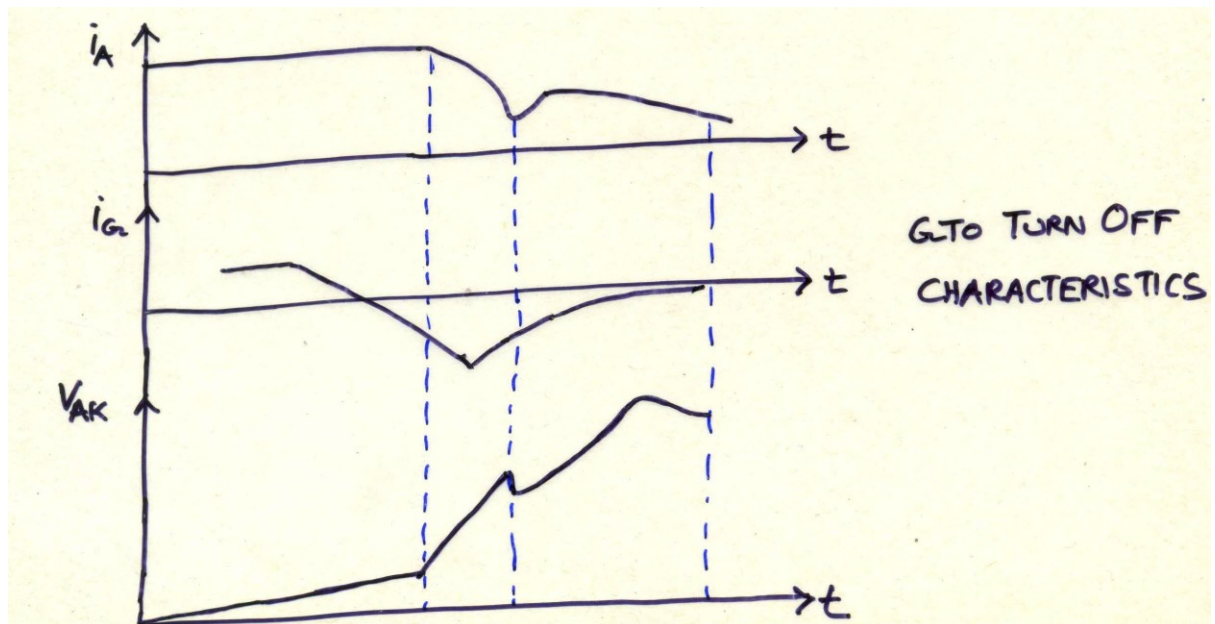
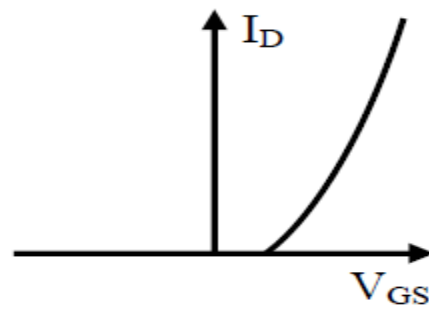
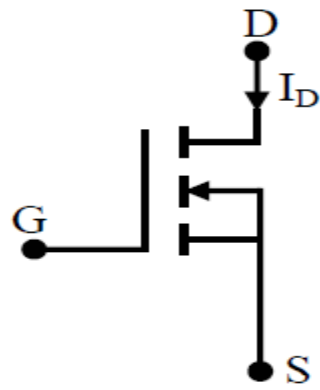


Fig. 5.3: Steady state characteristics of a GTO
 (a) Output characteristics; (b) Gate characteristics.



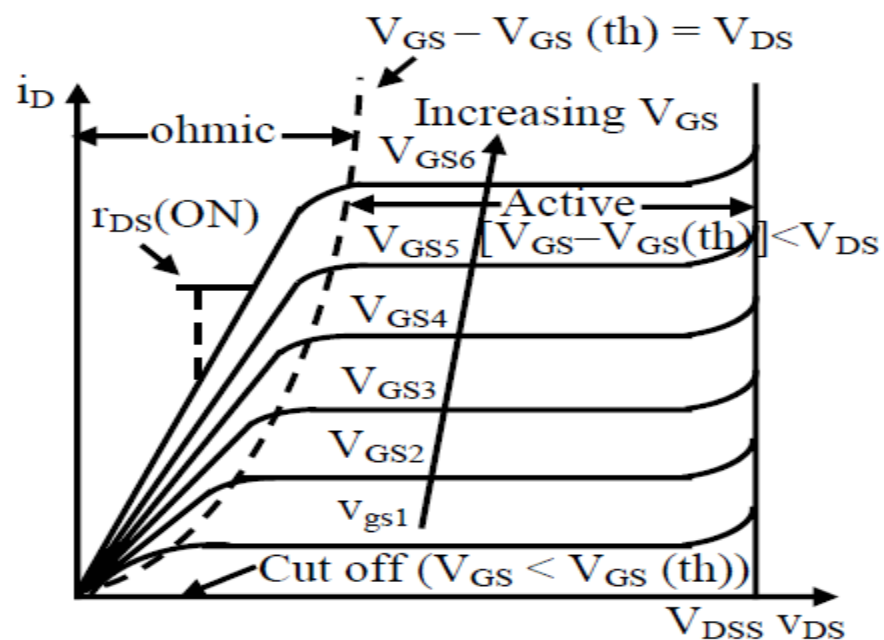


Schematic construction of a power MOSFET



n-channel enhancement
type MOSFET

Circuit symbols and transfer characteristics



Output i-v characteristics of a Power MOSFET

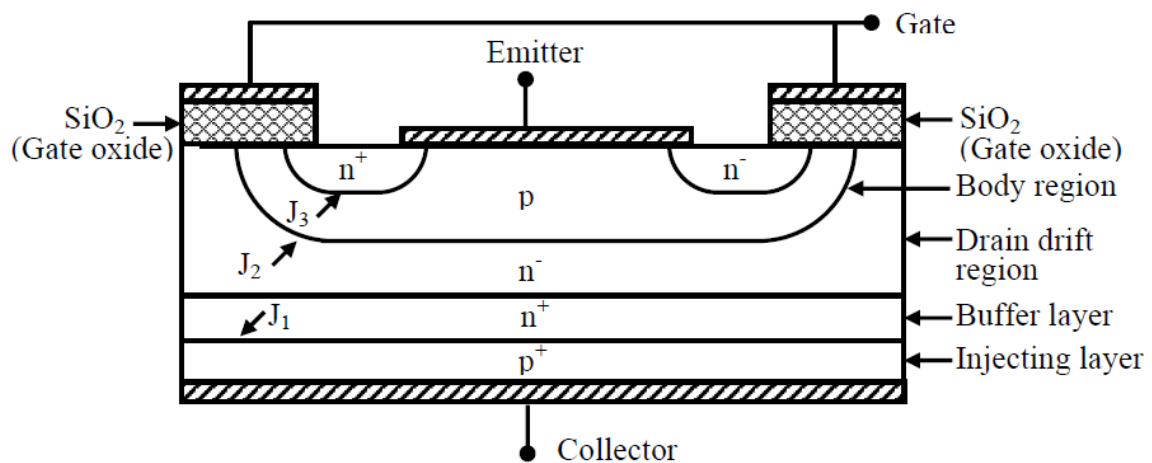
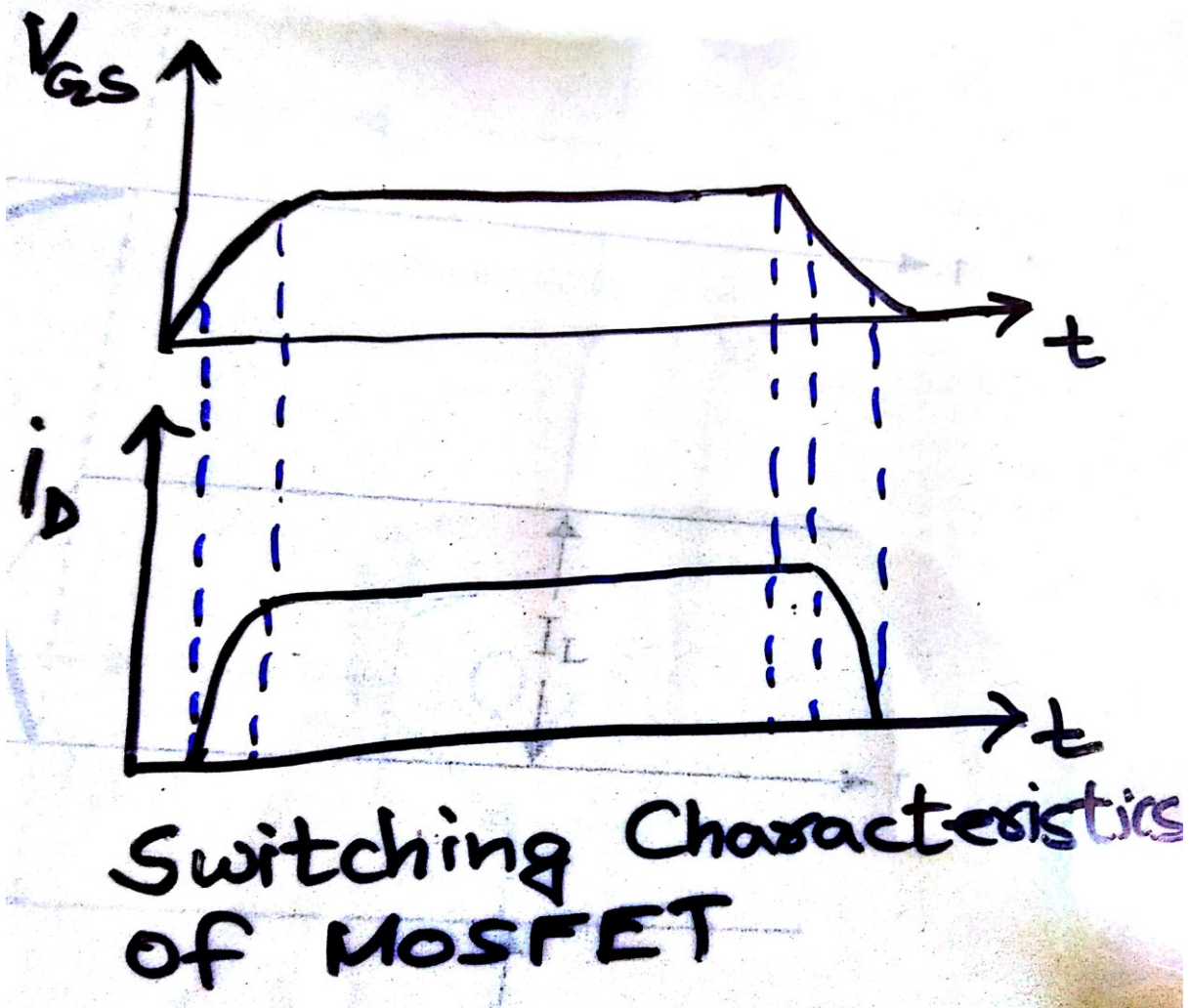


Fig. 7.1: Vertical cross section of an IGBT cell.



Fig. 7.3: Circuit symbol of an IGBT.

(a) Circuit symbol.

(b) Photograph.

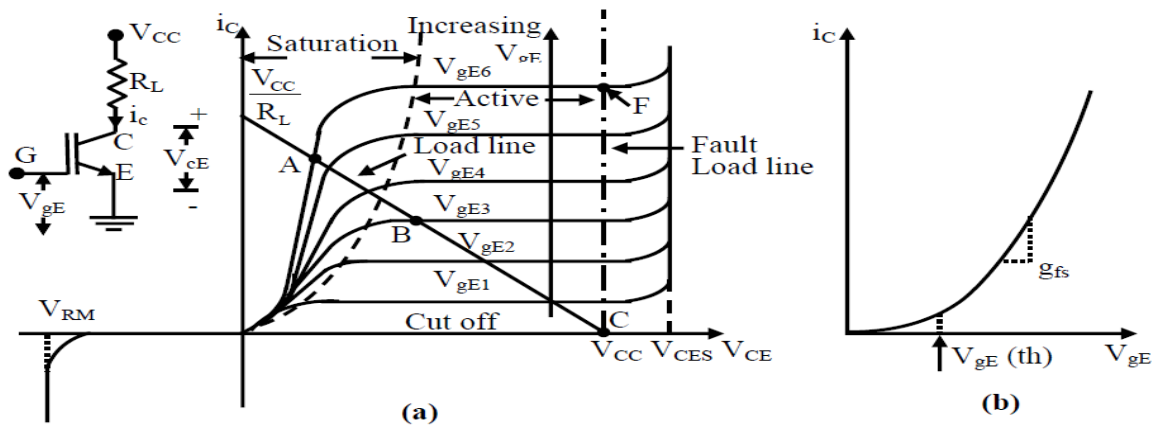


Fig. 7.4: Static characteristics of an IGBT

(a) Output characteristics; (b) Transfer characteristics

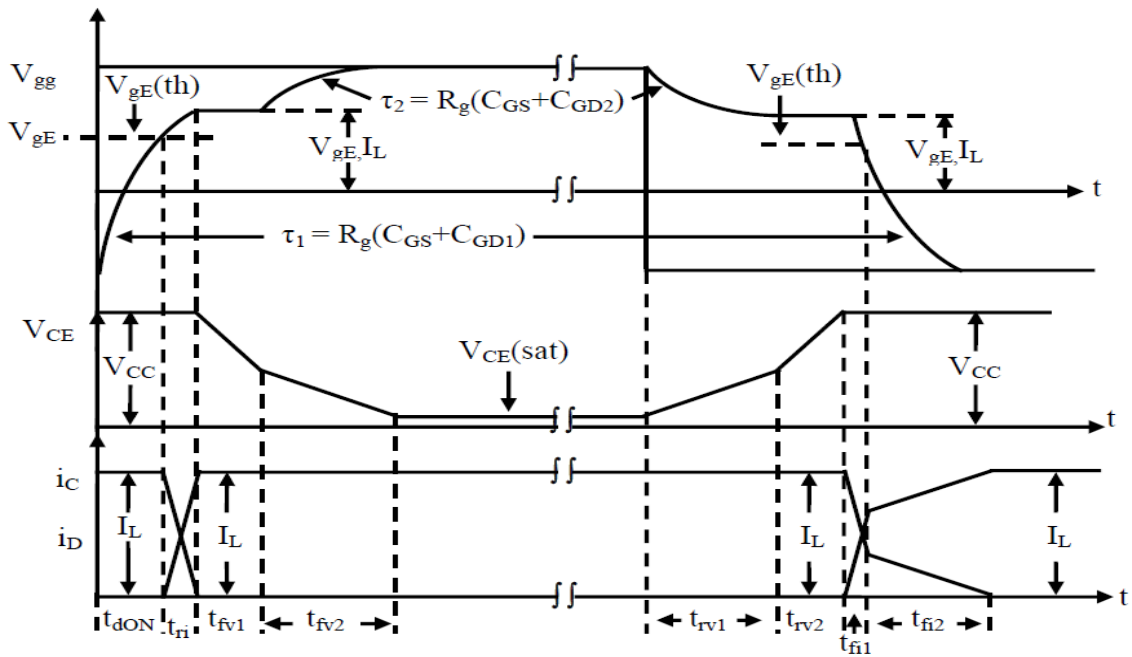


Fig. 7.6: Switching waveforms of an IGBT.

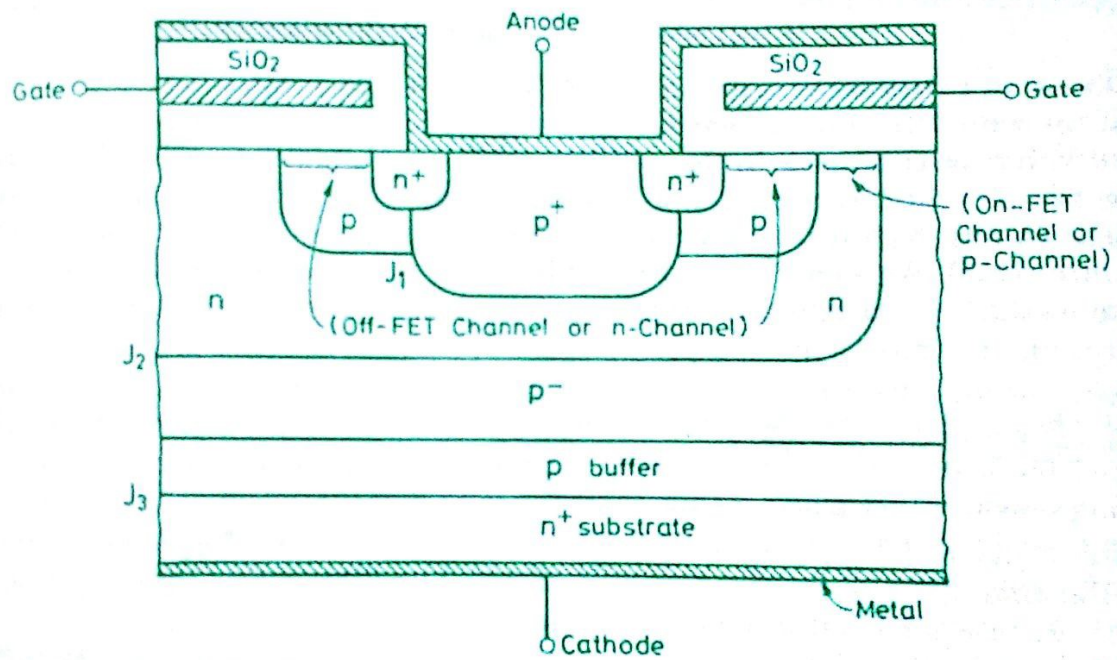


Fig. 2.27. Basic structure of an MCT.

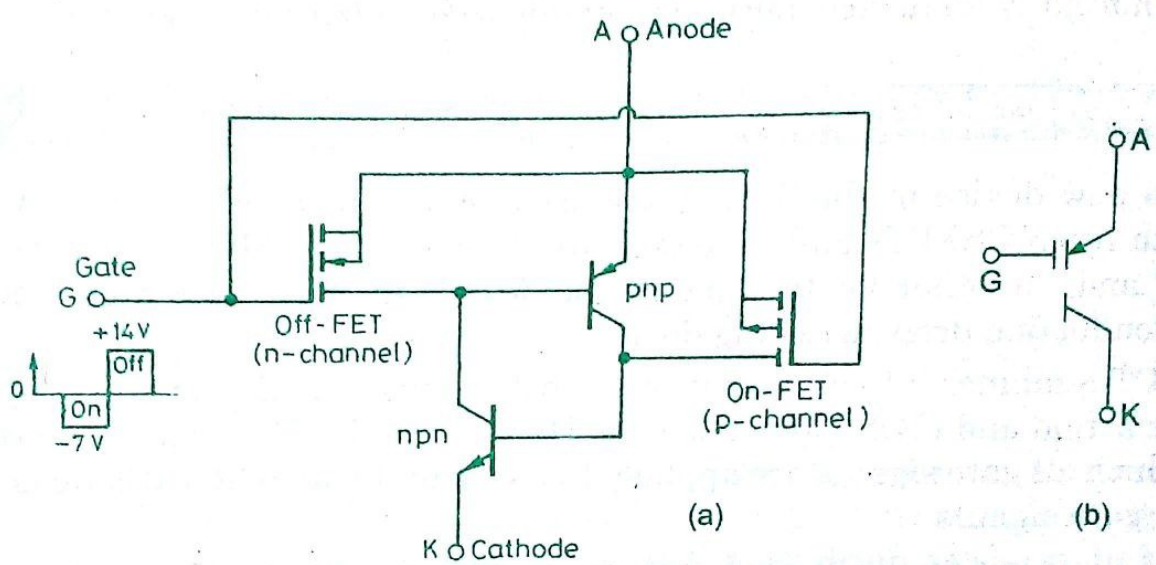


Fig. 2.28. MCT (a) equivalent circuit and (b) circuit symbol.

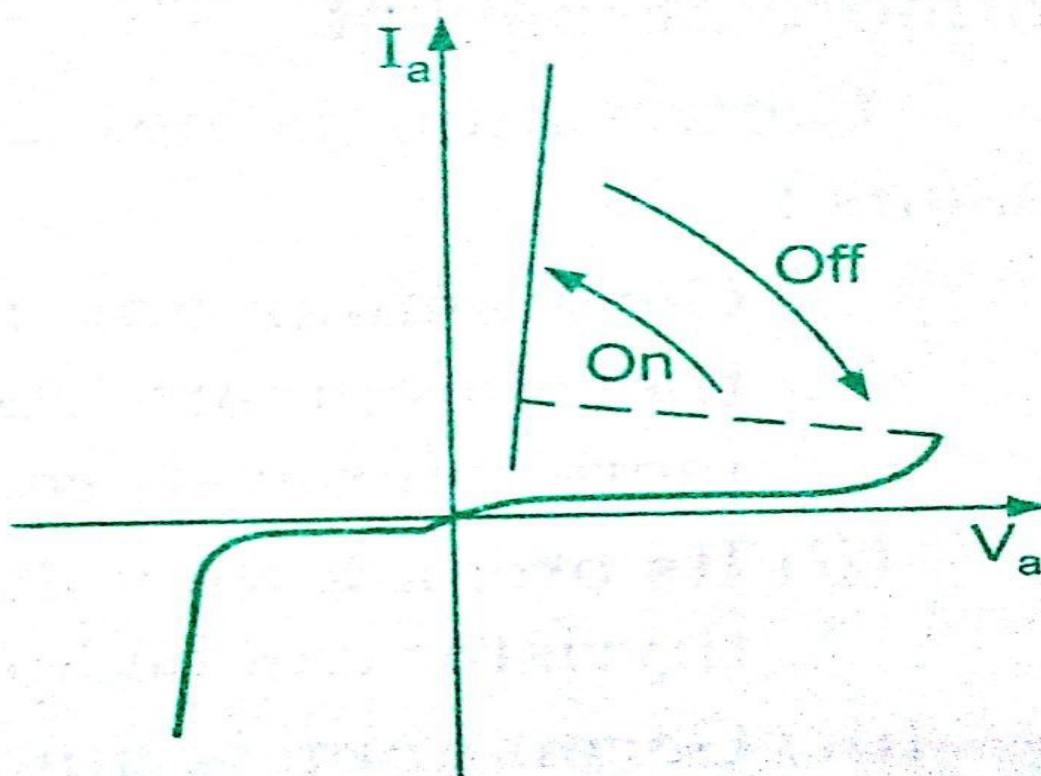
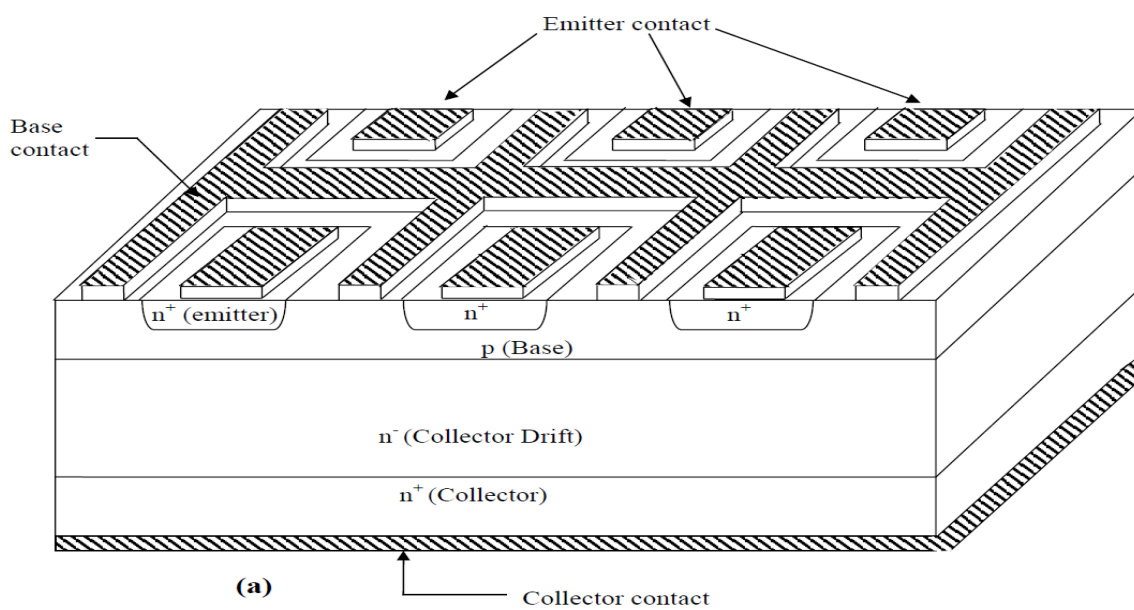


Fig. 2.29. Static I-V characteristics of an MCT.



Power Bipolar Junction Transistor (BJT)

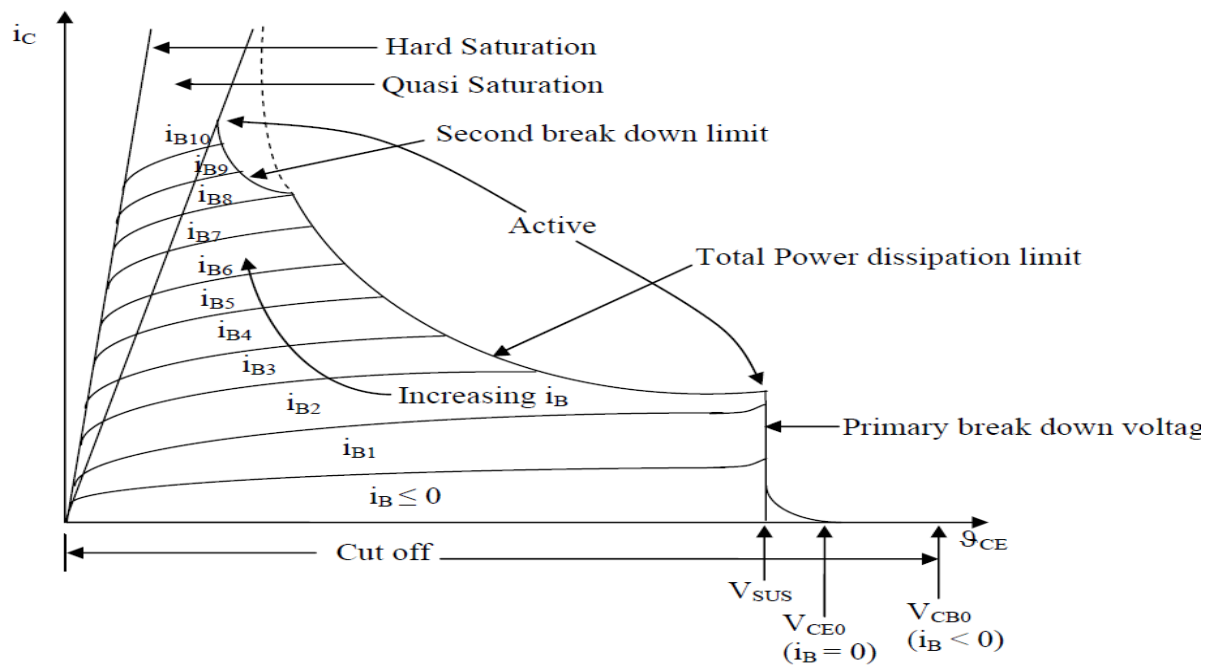


Fig. 3.4 Output ($i_c - v_{CE}$) characteristics of an n – p – n type Power Transistor

